

Template Curriculum Vitae JEROEN DEGROOTE
(MAX. 4 pages)

Research Statement: Data-driven animal nutrition – shaping the future of sustainable pig and poultry production

- **Short CV**

Jeroen Degroote is now Assistant Professor at the Department of Animal Sciences and Aquatic Ecology, Ghent University (31/08/2023-present, 1.0 FTE). He obtained a PhD in Applied Biological Sciences at Ghent University in 2019 and an MSc in Applied Biosciences: Plant and Animal Production in 2011. His previous positions were Doctor-Assistant/Post-doctoral Fellow at Ghent University (01/10/2018-01/09/2023, 1.0 FTE), Assistant and PhD Student at Ghent University (01/10/2014-30/09/2019, 1.0 FTE), PhD Student at Ghent University (01/10/2013-30/09/2014, 1.0 FTE), and Scientific Researcher at University College Ghent (01/10/2011-01/10/2014, 1.0 FTE).

His current academic work is divided between research, teaching, and service. Research and supervision take about 60% FTE. He supervises PhD students and thesis students working on pig nutrition, gut development, feed intake, protein efficiency, and sensor-assisted monitoring. Teaching takes about 30% FTE. Since 2023, he has been lecturer-in-charge for Animal Production Systems, Nutrition of Pigs, and Integrated Exercises: Crops, Animals and Agricultural Farm, and co-lecturer for Monitoring Systems in Agriculture and Animal Nutrition. Scientific and institutional service takes about 10% FTE. He is a member of the Department Board (2017-present) and the Ethical Committee for Animal Experimentation at Ghent University (2019-present).

- **Career path**

Jeroen Degroote leads the research group for pigs and poultry at LANUPRO, which is a dedicated team of PhD students, post-doc mentors, and specialized technicians who provide professional support in lab analysis and animal studies. His group studies nutritional interventions in pigs and poultry to improve animal health, promote efficient utilization of natural resources, and reduce environmental impact. This involves multidisciplinary research in digestive physiology and quantitative animal nutrition. Our research vision is to expand concepts in animal nutrition and digestive physiology and integrate this knowledge with modern technology in data management and analysis. This includes utilizing machine learning, decision-making tools, decentralized data storage, and advancements in data semantics. In addition to these fundamental projects, we routinely conduct contract research services for the industry. This involves assessing the nutritional value of novel feed ingredients and evaluating the potential effects of feed additives or alternative feed formulations. These services also include EFSA-compliant registration trials, delivering evidence that has supported the market authorization of feed additives such as digestive enzymes (e.g. phytase) or probiotics (e.g. *Bacillus subtilis*). This unique blend of mechanistic research and applied expertise positions the team to translate new nutritional strategies into commercially viable, environmentally sustainable solutions for pigs and poultry nutrition.

Career breaks

No career breaks

Five main publications and/or achievements

- S. Millet, J. Degroote, J. Teunen, S. Goethals and N. Everaert. "Review: Optimising Health, Welfare and Performance in Weaned Piglets, Striking the Ideal Balance between Amino Acids, Energy, and Protein." *animal: The International Journal of Animal Biosciences*, 20, article 101811, 2026. DOI: <https://doi.org/10.1016/j.animal.2026.101811>. Impact factor: 4.2. Peer reviewed: yes.
- Renée De Baets, Sofie Van Nerom, Kobe Buyse, Gunther Antonissen, Jeroen Degroote and Evelyne Delezie. "Testing Low Dietary Crude Protein and High Fat Levels as a Strategy to Mitigate Heat Stress in Broilers." *Journal of Animal Science and Biotechnology*, 16(1), article 175, 2025. DOI: <https://doi.org/10.1186/s40104-025-01297-4>. Impact factor: 6.5. Peer reviewed: yes.
- Chengcheng Li, Stijn Heirbaut, Joris Michiels, Yuhuang Hou, Anneke Ovyne, Noémie Van Noten, Elout Van Liefferinge and Jeroen Degroote. "Integrative Clustering by Colonic Volatile Fatty Acids Reveals Microbiota and Host Physiological Variation in Weaned Piglets." *Journal of Animal Science and Biotechnology*, 17(1), article 39, 2026. DOI: <https://doi.org/10.1186/s40104-026-01353-7>. Impact factor: 6.5. Peer reviewed: yes.
- S. Noor, J. Degroote, G. Van Schaik, B. Pardon, C. Faverjon, C. Delavenne and M. Hostens. "A Systematic Review on the Role of Livestock Ontologies in Animal Health Management and Disease Surveillance: A PRISMA 2020 and AI-Assisted Screening Approach." *Smart Agricultural Technology*, 14, article 101977, 2026. DOI: <https://doi.org/10.1016/j.atech.2026.101977>. Impact factor: 5.7. Peer reviewed: yes.
- T. Van De Putte, C. Van Kerschaver, M. Hostens and J. Degroote. "Investigating the Development of Piglet Feeding Behaviour during the Immediate Postweaning Phase Using Computer Vision." *animal: The International Journal of Animal Biosciences*, 19(6), article 101524, 2025. DOI: <https://doi.org/10.1016/j.animal.2025.101524>. Impact factor: 4.2. Peer reviewed: yes. *Computer Vision.* *Animal* 19, no. 6 (Jun 2025). <https://dx.doi.org/10.1016/j.animal.2025.101524>.

Other scientific output and impact

Jeroen Degroote has contributed to international scientific exchange through invited keynote lectures and conference organization. He was invited keynote speaker at ESPN 2025 in Maastricht, where he presented on nutritional management of heat-stressed broilers. He also gave an invited lecture at the 2026 BOKU Symposium on precision nutrition and protein feeding strategies in weaned piglets. In June 2026, he hosts the EAAP AI4Animal Science Conference at Ghent University.

Jeroen Degroote served as a promoter for 2 PhD theses: Yuhuang Hou (CSC scholarship, China) and Herinda Pertiwi (Grant from Airlangga University, Indonesia). He was also a co-promoter for 2 theses: Céline Van Kerschaver (VLAIO-LA, Belgium) and Renée De Baets (VLAIO-LA, Belgium). All these theses were successfully completed during the past 3 years. Currently, he is hosting 5 resident PhD students: Chengcheng Li (CSC scholarship, China), Yao Zhu (CSC scholarship, China), Thomas Van De Putte (BOF-‘starterskrediet’, Belgium), Ruicong Wang (Flanders’ Food cSB0) and Mario Van Daele (VLAIO-LA, Belgium). Additionally, He is promoting 3 PhD students; Yajing Wang (Flanders’ Food cSB0), Renée Kelem (Flanders’ Food cSB0) and Saba Noor (H2020, Europe), who are part of project collaborations outside but are not resident researchers at our group. One post-doc, Stijn Heirbaut (UGent postdoc mandate, Belgium), is assisting in mentoring these students. The team furthermore contains 3 FTE of technical staff for teaching, lab work and running animal trials.

Our international collaborations currently include contacts in China, which have so far resulted in careful selection of incoming CSC students by Prof. Xian-Ren Jiang, Prof. Honglin Yan, Prof. Wen Yao, and Prof. Ding Hongbiao. We also receive regular visits from Chinese research groups, such as the Institute of Feed Research at the Chinese Academy of Agricultural Sciences (CAAS).

List the representative and substantial fellowships, projects and any other kind of research grants you obtained within the five years preceding the submission date of this project application.

Within the five years preceding this application, I obtained or was involved in the following representative research funding related to pig nutrition, gut function, and data-driven animal nutrition:

- **Adisseo Research Grant, 2025-2028**
 - *COMPETENT PIGS: Continuous Monitoring of Protein Efficiency for Smarter Protein Nutrition in Weaned Piglets.*
 - This project focuses on precision protein nutrition in weaned pigs, with monitoring of feed intake, growth, and protein efficiency.
- **Flanders' Food cSBO, 2025-2029**
 - *NUTRITWIN: a digital twin for sustainable pig production*
 - This project supports data integration and data-driven animal nutrition in pig production. It is relevant to this application because it strengthens the use of digital tools and modelling for precision nutrition.
- **CSC Scholarship Program, Chengcheng Li, 2021-2025**
 - *Associations between developmental aspects of gut function and bacterial community by omics approaches in weaned piglets.*
 - This PhD project contributes to the understanding of gut development and microbial responses in piglets.
- **CSC Scholarship Program, Yao Zhu, 2022-2026**
 - *Continuous monitoring of protein efficiency for smarter animal nutrition in weaned piglets.*
 - This PhD project is closely linked to precision feeding and protein efficiency in pigs.

Together, these projects combine animal experiments, gut-health assessment, feed intake monitoring, and data-driven interpretation. This provides a useful basis for the current application, which also focuses on improving pig nutrition through more precise and better-informed feeding strategies.